

Day 1: Power BI Student Activity

Activity Structure

Time	Activity
0-5 min	Understand the business problem & data
5-30 min	Use Case 1 - Retail Sales Dashboard
30-55 min	Use Case 2 - HR Employee Analytics
55-60 min	Present findings / trainer review

USE CASE 1 – Retail Sales Performance Dashboard

25 minutes

Business Scenario

A retail company sells products across different regions and wants management to understand:

- Which products generate the most revenue?
- Which categories perform best?
- Which regions are performing well?
- How are sales changing over time?
- Which products are profitable?
- Can management interactively filter the dashboard?

Dataset

Given **3 tables**.

Table 1 - Sales

OrderID	OrderDate	ProductID	CustomerID	Region	Sales	Quantity	Profit
0001	05-Jan-26	P01	C01	South	50000	2	8000

OrderID	OrderDate	ProductID	CustomerID	Region	Sales	Quantity	Profit
0002	10-Jan-26	P02	C02	West	30000	3	5000
0003	15-Feb-26	P03	C03	North	45000	5	7000
0004	20-Feb-26	P01	C04	South	60000	3	10000
0005	05-Mar-26	P04	C05	East	25000	4	4000
0006	18-Mar-26	P02	C01	South	35000	4	6000
0007	10-Apr-26	P03	C06	West	55000	6	9000
0008	25-Apr-26	P04	C07	North	20000	3	3000
0009	05-May-26	P01	C08	East	70000	4	12000
0010	20-May-26	P02	C09	South	40000	5	7000

Table 2 - Product

ProductID	Product	Category
P01	Laptop	Technology
P02	Monitor	Technology
P03	Office Chair	Furniture
P04	Desk	Furniture

Table 3 - Customer

CustomerID	Customer	State	City
C01	ABC Ltd	Tamil Nadu	Chennai
C02	XYZ Ltd	Maharashtra	Mumbai
C03	PQR Ltd	Delhi	Delhi
C04	LMN Ltd	Tamil Nadu	Chennai
C05	DEF Ltd	West Bengal	Kolkata
C06	GHI Ltd	Karnataka	Bengaluru
C07	JKL Ltd	Delhi	Delhi
C08	MNO Ltd	Maharashtra	Mumbai
C09	STU Ltd	Tamil Nadu	Chennai

Student Tasks

Task 1 - Load Data

2 minutes

Load all three tables into Power BI.

Steps:

1. **Home** → **Get Data** → **Excel**
 2. Select tables
 3. **Load**
-

Task 2 - Create Relationships

4 minutes

Create:

```
Product [ProductID]
    1
    |
    *
```

```
Sales [ProductID]
```

and:

```
Customer [CustomerID]
    1
    |
    *
```

```
Sales [CustomerID]
```

Should use **Model View**.

Task 3 - Create KPI Cards

3 minutes

Create three cards:

- Total Sales
- Total Profit

- Total Quantity

Example:

TOTAL SALES

₹410K

TOTAL PROFIT

₹71K

TOTAL QUANTITY

36

Task 4 - Sales by Category

3 minutes

Create a **Clustered Column Chart**.

Field	Placement
Category	Axis
Sales	Values

Title: Sales by Category

Task 5 - Monthly Sales Trend

3 minutes

Create a **Line Chart**.

Field	Placement
OrderDate	X-axis
Sales	Y-axis

Show sales over time.

Task 6 - Geographic Analysis

3 minutes

Create a map.

Field	Placement
State	Location
Sales	Size/Values

should identify which state has the highest sales.

Task 7 - Add Slicers

3 minutes

Add slicers for:

- **Region**
- **Category**

should click different values and observe how the dashboard changes.

Task 8 - Business Questions

2 minutes

It should answer:

1. Which category has the highest sales?
 2. Which region performs best?
 3. Which product generates the highest profit?
 4. Which state has the highest sales?
 5. What happens when **Technology** is selected?
 6. Does the monthly sales trend increase or decrease?
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USE CASE 2 - Employee / HR Analytics Dashboard

25 minutes

This second use case gives a completely different business scenario and makes them apply the same Power BI concepts to a new domain.

Business Scenario

An organization wants HR managers to understand its workforce.

Management wants to know:

- How many employees are there?
- Which department has the most employees?
- What is the average salary?
- Which locations have the highest workforce?
- How many employees have left?
- Which department has the highest attrition?

Dataset

Given **2 tables**.

Table 1 - Employee

EmployeeID	Name	DepartmentID	Location	Gender	Salary	JoiningDate	Attrition
E001	Arun	D01	Chennai	M	55000	10-Jan-23	No
E002	Priya	D02	Bengaluru	F	65000	15-Feb-22	No
E003	Ravi	D01	Chennai	M	48000	20-Mar-24	Yes
E004	Divya	D03	Mumbai	F	72000	05-Apr-21	No
E005	Karthik	D02	Bengaluru	M	60000	12-May-23	Yes
E006	Meena	D03	Mumbai	F	75000	18-Jun-22	No
E007	Suresh	D01	Chennai	M	50000	22-Jul-24	No
E008	Anitha	D02	Hyderabad	F	58000	01-Aug-23	No
E009	Bala	D03	Mumbai	M	70000	15-Sep-22	Yes
E010	Lakshmi	D01	Hyderabad	F	52000	10-Oct-23	No
E011	Vijay	D02	Bengaluru	M	62000	25-Nov-21	No
E012	Nisha	D03	Mumbai	F	68000	05-Dec-24	No

Table 2 - Department

DepartmentID	Department
D01	Finance
D02	IT

DepartmentID	Department
D03	HR

Student Tasks

Task 1 - Load the Tables

2 minutes

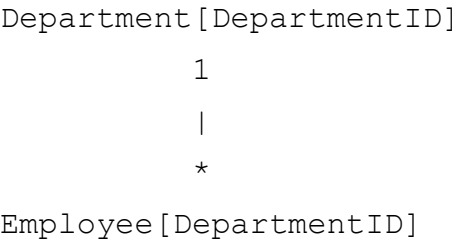
Load:

- Employee
- Department

Task 2 - Create Relationship

3 minutes

Create:



Task 3 - KPI Cards

4 minutes

Create cards for:

Card	Measure / Logic
Total Employees	Count of EmployeeID
Average Salary	Average of Salary
Employees Left	Count of employees where Attrition = Yes

Task 4 - Employees by Department

3 minutes

Create a **Bar Chart**.

Field	Placement
Department	Axis
Count of EmployeeID	Values

Question: Which department has the highest number of employees?

Task 5 - Salary Analysis

3 minutes

Create a column/bar chart showing **Average Salary by Department**.

Field	Placement
Department	Axis
Average Salary	Values

Task 6 - Geographic Analysis

3 minutes

Create a map.

Field	Placement
Location	Location
Employee Count	Size

Question: Which city has the largest employee population?

Task 7 - Attrition Analysis

3 minutes

Create a **Donut Chart**.

Field	Placement
Attrition	Legend
Employee Count	Values

Question: What percentage of employees have left?

Task 8 - Slicers

4 minutes

Add slicers for:

- **Department**
- **Gender**
- **Location**

Ask is to test:

- What happens to the average salary when you select **IT**?
 - What happens when you select **Female**?
-

Final Dashboard Expected

Students should create something approximately like:

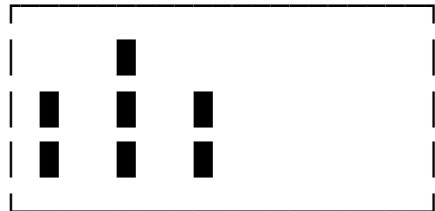
=====

EMPLOYEE ANALYTICS

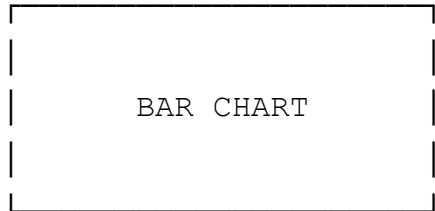
=====

[Employees] [Avg Salary] [Employees Left]

Employees by Department



Average Salary by Department



Department

IT
HR
Finance

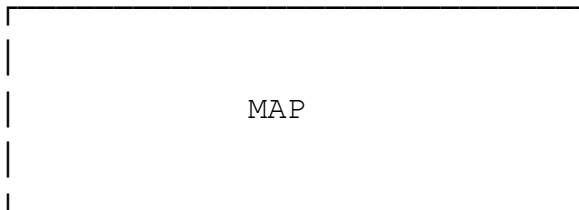
Gender

Male
Female

Location

Chennai
Mumbai
Bengaluru

Employee Location Map



Day 1Trainer Evaluation

At the end, don't just check whether the dashboard looks good. Ask is to **explain the business decisions**.

Ask each group/student:

1. Why did you use a column chart here?

Expected:

"Because we are comparing categories."

2. Why did you use a line chart?

Expected:

"To identify a trend over time."

3. Why did you create relationships?

Expected:

"To allow data from different tables to work together correctly."

4. What is the difference between Append and Merge?

Expected:

Append adds rows; Merge adds columns using matching fields.

5. What does the slicer do?

Expected:

"It allows users to interactively filter the report."

6. What happens when you select a category?

Expected:

"The connected visuals respond based on the selected filter."
